Evolution of Dmc1 Homologs Across Metamonada

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Background, Rationale, and Aims

Background

Metamonada is a diverse group of anaerobic protists with reduced or absent mitochondria [3]. Their rapid evolution and unique biology make them an important system for studying how DNA repair mechanisms may influence genome evolution.

Rationale

Dmc1 plays a key role in meiotic homologous recombination [1]. Previous work has shown Dmc1 duplications in Fornicata associated with loss of Rad51A [2], but the broader evolutionary patterns across Metamonada and the relationship to Hop2 and Mnd1 remain unclear.

Aims

- Characterize the phylogenetic distribution of Dmc1 across Metamonada.
- Compare Dmc1 evolutionary patterns to Hop2 and Mnd1.
- Investigate potential gene duplications and losses in the context of homologous recombination evolution.

Methods

Retrieve sequences

Orthogroup OG6_100483 (Eukphylo)

Inspect and prune

Remove redundant sequences (Taxus) 1,222 → 1,131 sequences (91 removed)

Align sequences

MAFFT (alignment)

Trim alignment

trimAl (trimmed alignment)

Build phylogeny

IQ-TREE 1,000 UFBoot (ML phylogeny)

Visualize & format

FigTree Adobe Illustrator

Dmc1 Phylogeny

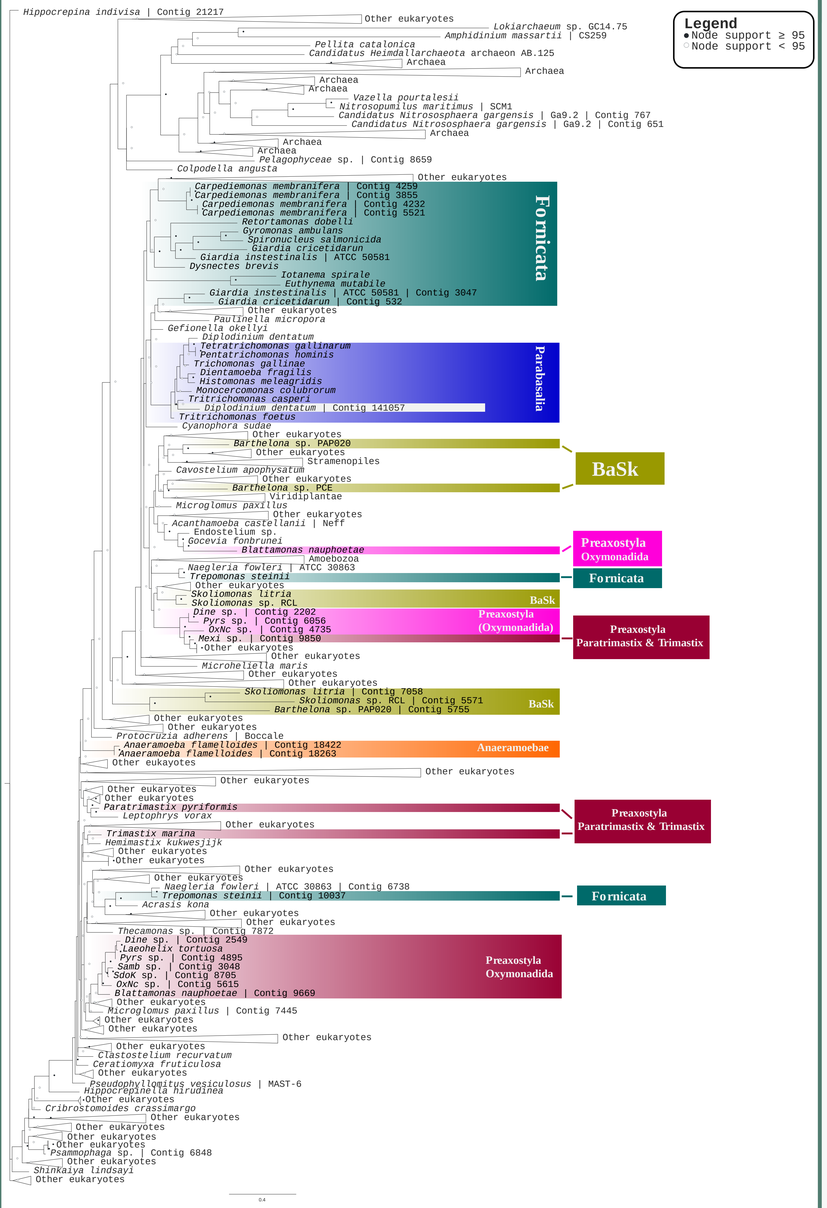


Figure 1. Detailed phylogeny of Dmc1 homologs across Metamonada. Metamonad sequences are highlighted by lineage, showing their distribution across multiple regions of the 1,131-sequence maximum-likelihood phylogeny. UFBoot values ≥95 are shown.

Results

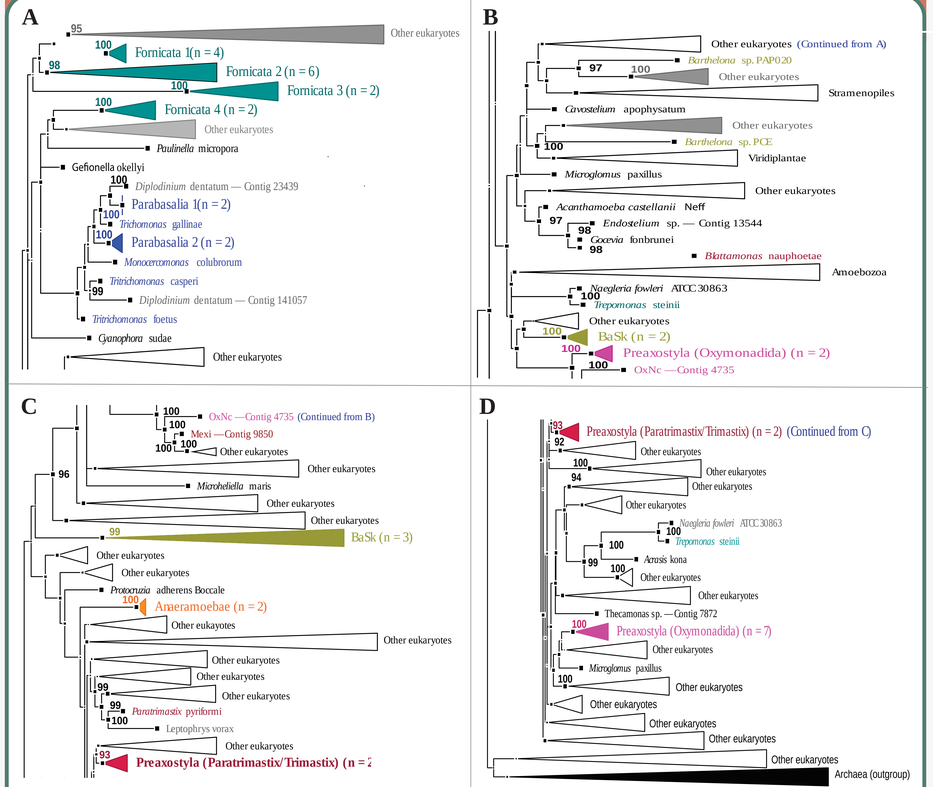


Figure 2. Distribution of Dmc1 homologs across Metamonada. Metamonad lineages occur across multiple regions of the Dmc1 phylogeny. Multiple Dmc1 copies were observed in Fornicata, BaSk, and Preaxostyla, while Anaeramoebae was concentrated in a single region. UFBoot ≥95 is shown.

Conclusions

- Dmc1 homologs are broadly distributed across Metamonada, occurring in multiple distinct regions of the phylogeny.
- Multiple Dmc1 copies were observed beyond Fornicata, including in BaSk and Preaxostyla.
- These patterns suggest a more complex evolutionary history than a single Fornicata-specific Dmc1 duplication.
- Additional taxon and sequence sampling is needed to clarify Dmc1 duplication and loss events across Metamonada.

References

1. Kagawa W, Kurumizaka H. 2010. From meiosis to postmeiotic events: Uncovering the molecular roles of the meiosis-specific recombinase Dmc1. The FEBS Journal 277:590–598.
2. Salas-Leiva DE, et al. 2021. Genomic analysis finds no evidence of canonical eukaryotic DNA processing complexes in a free-living protist. Nature Communications 12:6003.
3. Williams SK, et al. 2024. Extreme mitochondrial reduction in a novel group of free-living metamonads. Nature Communications 15:6805.

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